Getting Started with epiRomics

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Abstract

Summary epiRomics is an R package designed to integrate multi-omics data in order to identify and visualize enhancer regions alongside gene expression and other epigenomic modifications. Regulatory network analysis can be done using combinatory approaches to infer regions of significance such as enhancers, when combining ChIP and histone data. Downstream analysis can identify co-occurrence of these regions of interest with other user-supplied data, such as chromatin availability or gene expression. Finally, this package allows for results to be visualized at high resolution in a stand-alone browser.

Availability and Implementation epiRomics is released under the Artistic-2.0 License. The source code and documents are freely available through GitHub (https://github.com/Huising-Lab/epiRomics).

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Supplementary information Supplementary data and methods are available online on *bioRxiv* or *GitHub*.

Competing Interest Statement The authors have declared no competing interest.

Citation

If you use epiRomics in published research, please cite:

Mawla AM, van der Meulen T, Huising MO. **Chromatin accessibility differences between alpha, beta, and delta cells identifies common and cell type-specific enhancers.** BMC Genomics. 2023;24:202. https://doi.org/10.1186/s12864-023-09293-6

Mawla AM & Huising MO. **epiRomics: a multi-omics R package to identify and visualize enhancers.** bioRxiv. 2021. https://doi.org/10.1101/2021.08.19.456732

Shiny epiRomics Showcase

Explore epiRomics results interactively through our companion Shiny web application:

Shiny epiRomics — Interactive browser for pancreatic islet enhancer and enhanceosome analysis results.

Example Data Setup

Note: epiRomics example data (~1.3 GB) is hosted remotely and must be downloaded once before running this vignette with live output. If the data is not yet cached, code chunks below will display the source code without execution.

```
## Run this once to download and cache example data
epiRomics_cache_data()
```

Data status: Example data is available. All code chunks will execute with live output.

Loading the epiRomics package

```
library(epiRomics)
#>
#>
#> epiRomics v0.99.0 - Epigenomic Analysis Package
#> Run epiRomics_cache_data() to download example datasets (~1.3 GB).
#> Citation: citation("epiRomics")
library(TxDb.Hsapiens.UCSC.hg38.knownGene)
#> Loading required package: GenomicFeatures
#> Loading required package: BiocGenerics
#> Loading required package: generics
#>
#> Attaching package: 'generics'
#> The following objects are masked from 'package:base':
#>
#>     as.difftime, as.factor, as.ordered, intersect, is.element,
#>     setdiff, setequal, union
#>
#> Attaching package: 'BiocGenerics'
#> The following objects are masked from 'package:stats':
#>
#>     IQR, mad, sd, var, xtabs
#> The following objects are masked from 'package:base':
#>
#>     anyDuplicated, aperm, append, as.data.frame, basename,
#>     cbind, colnames, dirname, do.call, duplicated, eval,
#>     evalq, Filter, Find, get, grep, grepl, is.unsorted,
#>     lapply, Map, mapply, match, mget, order, paste, pmax,
#>     pmax.int, pmin, pmin.int, Position, rank, rbind, Reduce,
#>     rownames, sapply, saveRDS, table, tapply, unique, unsplit,
#>     which.max, which.min
#> Loading required package: S4Vectors
#> Loading required package: stats4
#>
#> Attaching package: 'S4Vectors'
#> The following object is masked from 'package:utils':
#>
#>     findMatches
#> The following objects are masked from 'package:base':
```

```
#>
#>     expand.grid, I, unname
#> Loading required package: IRanges
#> Loading required package: Seqinfo
#> Loading required package: GenomicRanges
#> Loading required package: AnnotationDbi
#> Loading required package: Biobase
#> Welcome to Bioconductor
#>
#>     Vignettes contain introductory material; view with
#>     'browseVignettes()'. To cite Bioconductor, see
#>     'citation("Biobase")', and for packages
#>     'citation("pkgname")'.
library(org.Hs.eg.db)
```

Brief explanation of example data

This package includes example data delineating human pancreatic islet enhancers between alpha and beta cells.

Human pancreatic islet alpha and beta ATAC- and companion RNA-Seq data were retrieved from GEO accession GSE76268 (Ackermann et al., 2016).

ATAC samples were processed using the ENCODE-DCC ATAC sequencing pipeline, aligning to the hg38 build of the human genome (Harrow et al., 2012; Consortium, 2012; Davis et al., 2018).

Peak calls generated through the pipeline using MACS2 (Zhang et al., 2008) were analyzed downstream through the Bioconductor package DiffBind (Ross-Innes et al., 2012) to identify differentially enriched chromatin regions between the two cell types.

RNA samples were quality controlled using fastp (Chen et al., 2018), and aligned using STAR (Dobin et al., 2013) to hg38. Wiggle files produced by the STAR aligner were merged by cell type using UCSC command line tools.

BigWig files were subsetted using UCSC command line tools (Kent et al., 2010).

ChIP-seq peak calls for human pancreatic islet transcription factors Foxa2, MafB, Nkx2.2, Nkx6.1, and Pdx1 were retrieved from EMBL-EBI E-MTAB-1919 (Pasquali et al., 2014).

Histone modification peak calls for H3K27ac, H3K4me1, H3K27me3, H3K9me3, H3K4me3, H3K36me3 were retrieved from GEO GSE16256 (Bernstein et al., 2010), and H2A.Z from EMBL-EBI E-MTAB-1919 (Pasquali et al., 2014).

All peak calls retrieved as hg19 were lifted over to hg38 using the UCSC liftOver (Kent et al., 2002).

The FANTOM5 human enhancer database (Lizio et al., 2015), ultra-conserved non-coding elements from the UCNE database (Dimitrieva and Bucher, 2012), and the human islet regulome (Miguel-Escalada et al., 2019) were also retrieved and lifted to hg38.

Building the database

The database sheet CSV defines the input annotations. Each row specifies a BED file with its name, relative path, genome build, format, and type.

Required columns: **name**, **path**, **genome**, **format**, **type**. Genome must be in proper format (e.g., `mm10` or `hg38`). Type can be: `histone`, `methyl`, `SNP`, `chip`, or `functional`.

```r
example_epiRomics_Db_sheet <- read.csv(
  file.path(extdata_dir,
    "example_epiRomics_Db_sheet.csv"))
head(example_epiRomics_Db_sheet, 5)
#>       name                      path genome format    type
#> 1  h3k27ac  Histone/H3k27ac_hg38.bed   hg38    bed histone
#> 2  h3k4me1  Histone/H3K4me1_hg38.bed   hg38    bed histone
#> 3 h3k27me3 Histone/H3K27me3_hg38.bed   hg38    bed histone
#> 4  h3k9me3  Histone/H3K9me3_hg38.bed   hg38    bed histone
#> 5  h3k4me3  Histone/H3K4me3_hg38.bed   hg38    bed histone
```

`epiRomics_build_dB()` constructs the S4 database from the data sheet. The CSV uses relative paths; the `data_dir` parameter resolves them against the cache directory automatically.

```r
epiRomics_dB <- epiRomics_build_dB(
  epiRomics_db_file = file.path(extdata_dir,
    "example_epiRomics_Db_sheet.csv"),
  txdb_organism = paste0(
    "TxDb.Hsapiens.UCSC.hg38.knownGene::",
    "TxDb.Hsapiens.UCSC.hg38.knownGene"),
  epiRomics_genome = "hg38",
  epiRomics_organism = "org.Hs.eg.db",
  data_dir = extdata_dir)
#>
#> Attaching package: 'Biostrings'
#> The following object is masked from 'package:base':
#>
#>     strsplit
#> Building enhancers...
#> loading from cache
#> 'select()' returned 1:1 mapping between keys and columns
#> Building promoters...
#> Building 1to5kb upstream of TSS...
#> Building intergenic...
#> Building cds...
#> Building 5UTRs...
#> Building 3UTRs...
#> Building exons...
#> Building first exons...
#> Building introns...
#> Building intron exon boundaries...
#> Building exon intron boundaries...
#> Building CpG islands...
#> Building CpG shores...
#> Building CpG shelves...
#> Building inter-CpG-islands...
#> Building lncRNA transcripts...
#> loading from cache
```

Quick visualization with epiRomics_quick_view

Before running the full enhancer pipeline, epiRomics provides a zero-configuration entry point for the ultra-fast visualization of any gene locus with BigWig signal tracks. No database setup is required—just pass a gene symbol and BigWig file paths.

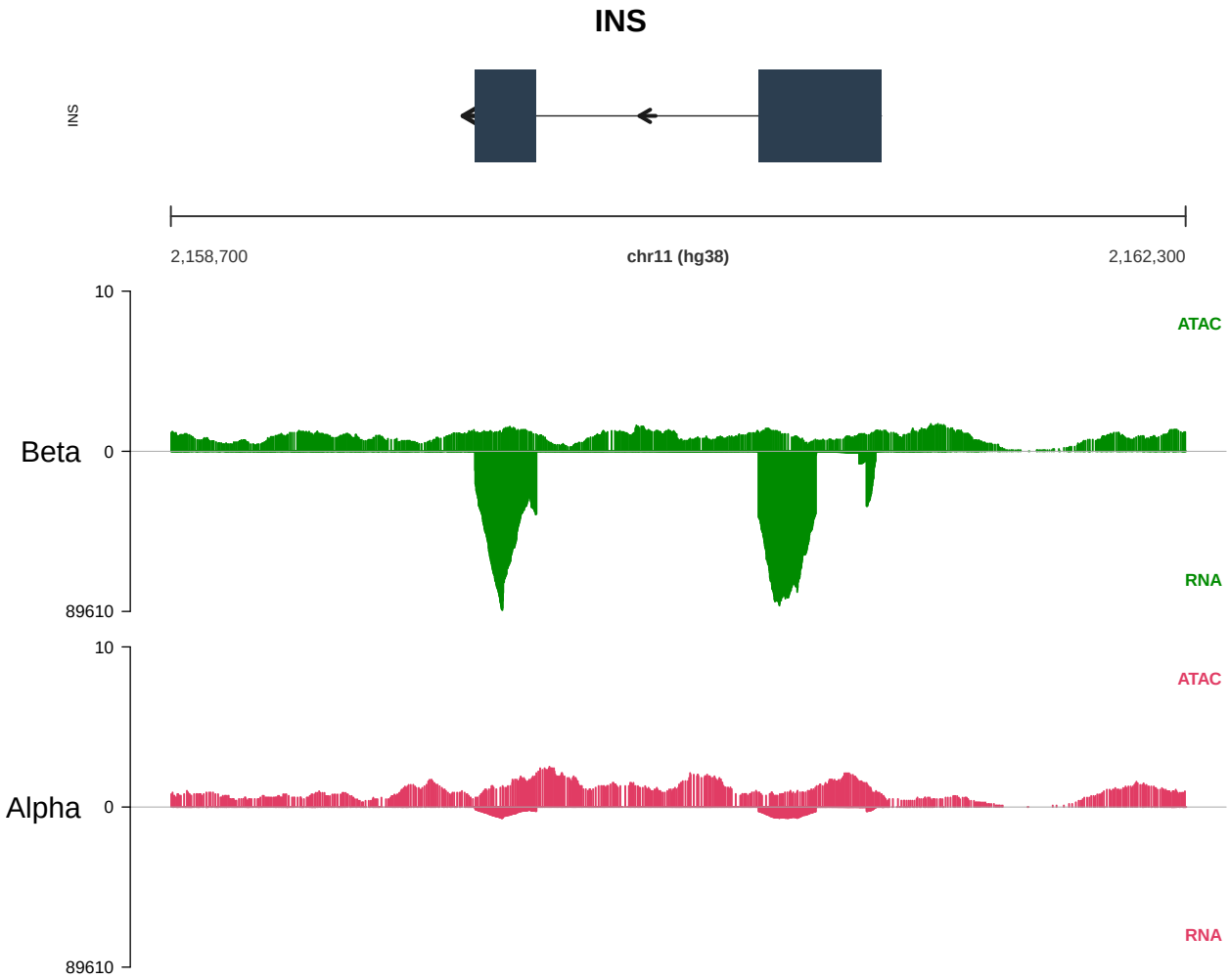
```r
## Load BigWig track connection sheet and resolve paths
epiRomics_track_connection <- read.csv(
  file.path(extdata_dir,
    "example_epiRomics_BW_sheet.csv"))
epiRomics_track_connection$path <- file.path(
  extdata_dir,
  epiRomics_track_connection$path)

## Reorder: Beta first, then Alpha
beta_idx <- grep("Beta", epiRomics_track_connection$name)
alpha_idx <- grep("Alpha", epiRomics_track_connection$name)
epiRomics_track_connection <- epiRomics_track_connection[
  c(beta_idx, alpha_idx), ]

## Named vector of BigWig paths and colors
bw_paths <- setNames(
  epiRomics_track_connection$path,
  epiRomics_track_connection$name)
bw_colors <- epiRomics_track_connection$color
```

The `mirror = TRUE` option displays ATAC and RNA signals in a mirrored layout for direct comparison between cell types.

Beta cell genes

INS

```r
epiRomics_quick_view("INS",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```

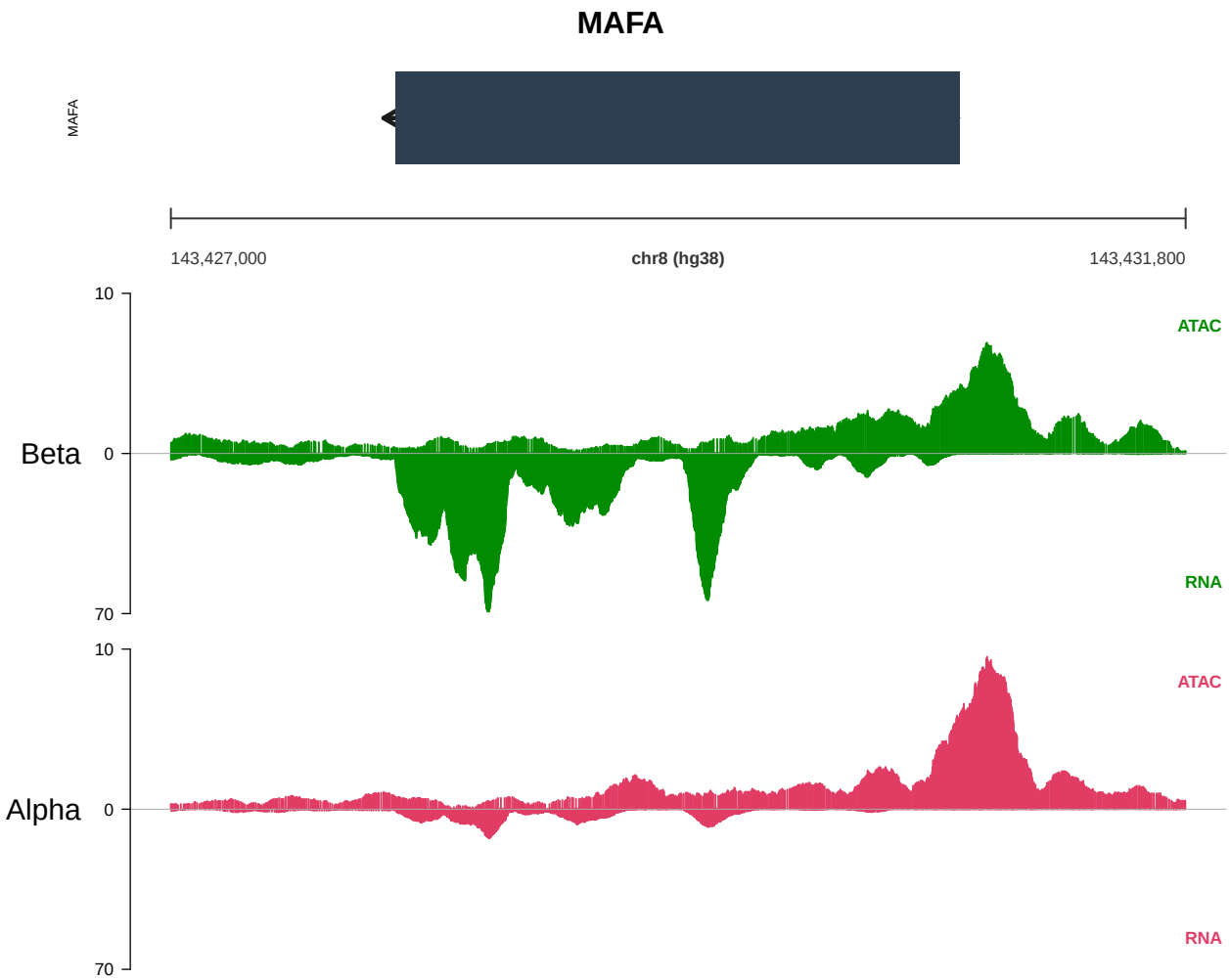


PDX1

```
epiRomics_quick_view("PDX1",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```

MAFA

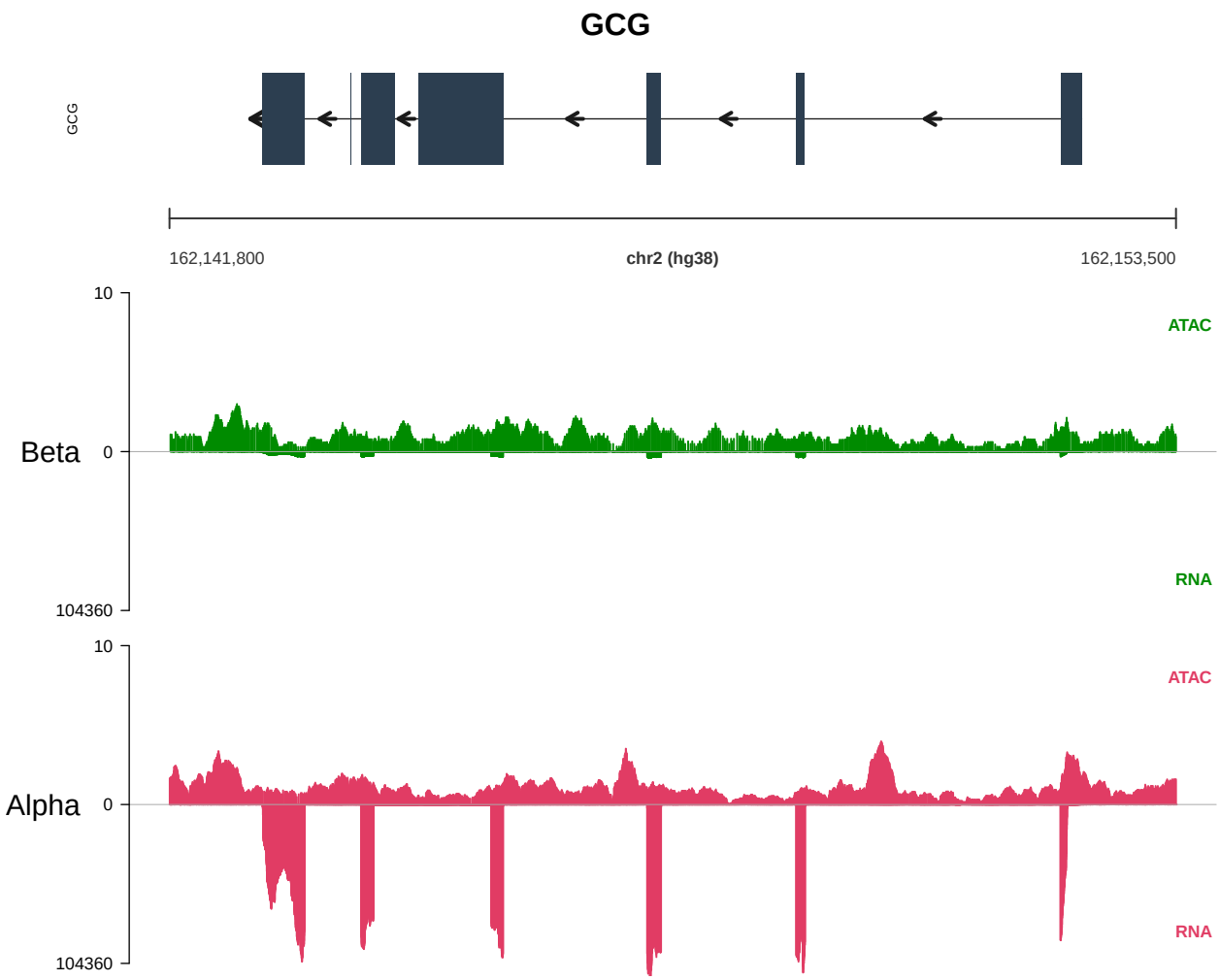
```
epiRomics_quick_view("MAFA",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```



Alpha cell genes

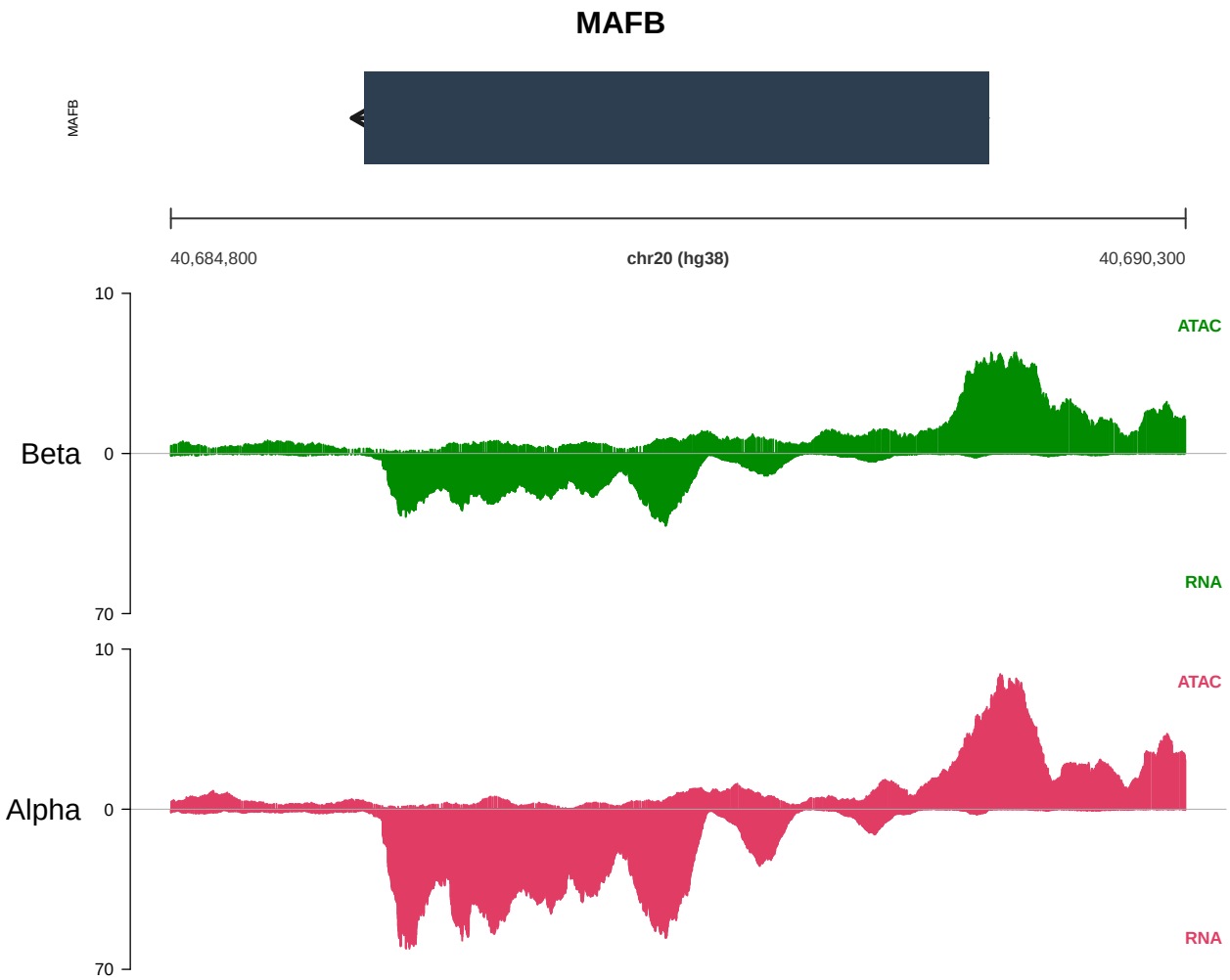
GCG

```
epiRomics_quick_view("GCG",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```



MAFB

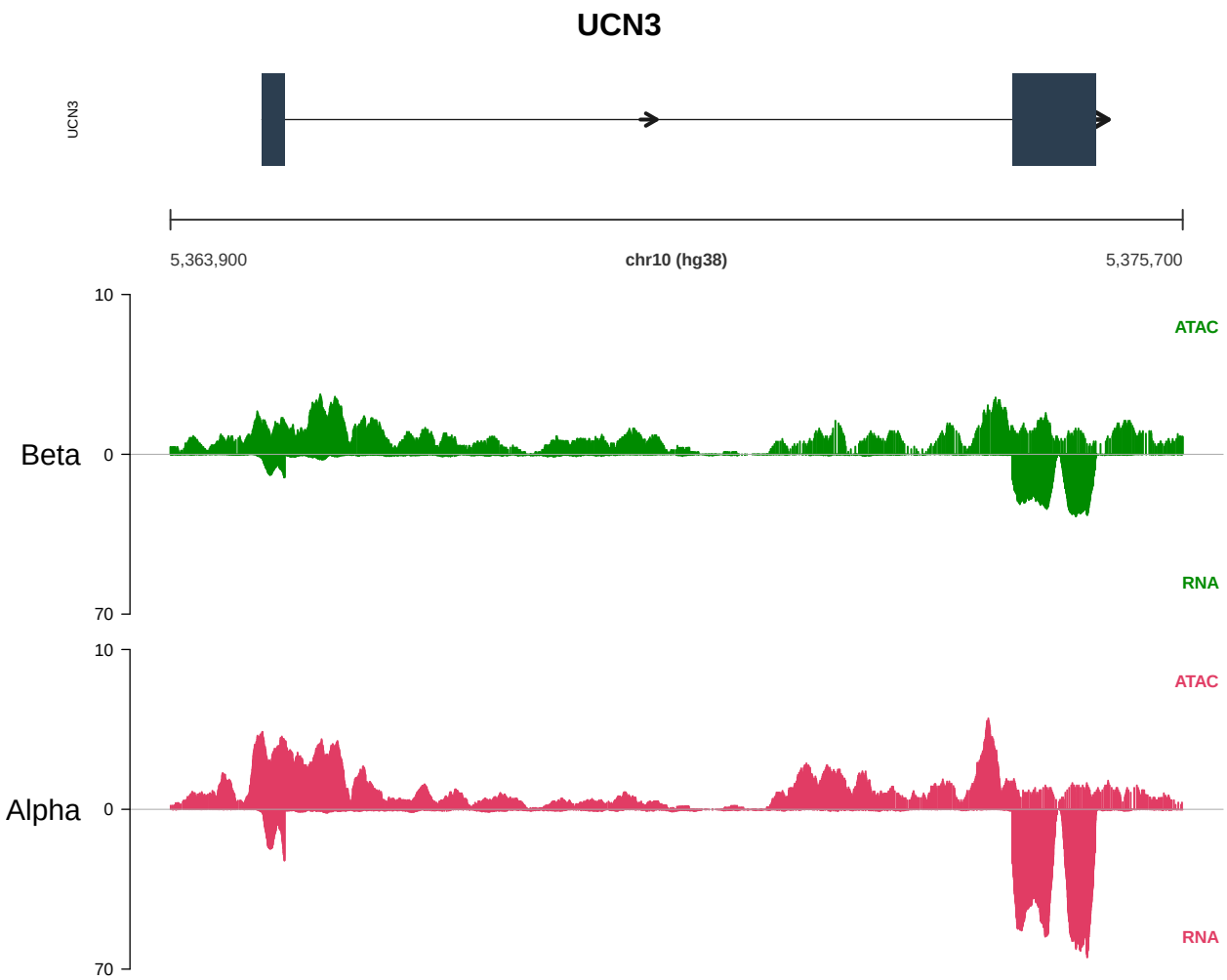
```
epiRomics_quick_view("MAFB",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```



Both Alpha and Beta Cell Genes

UCN3

```
epiRomics_quick_view("UCN3",
  bw_paths = bw_paths,
  colors = bw_colors,
  mirror = TRUE,
  genome = "hg38")
```



Identifying putative enhancers

There is a lot of flexibility for data exploration here. In this example, we search for putative enhancers using two histone marks known to co-occur at enhancer regions: H3K4me1 and H3K27ac.

```
epiRomics_putative_enhancers <-
  epiRomics_enhancers_co_marks(
    epiRomics_dB,
    epiRomics_histone_mark_1 = "h3k4me1",
    epiRomics_histone_mark_2 = "h3k27ac")

## First 5 putative enhancers
head(as.data.frame(
  epiRomics_putative_enhancers@annotations), 5)
#>   seqnames  start    end width strand
#> 1     chr1 816857 817332   476      *
#> 2     chr1 941074 941572   499      *
#> 3     chr1 941703 941976   274      *
#> 4     chr1 942153 942691   539      *
#> 5     chr1 942923 943021    99      *
```

Cross-referencing enhancer calls to databases

FANTOM Enhancer Database

Cross-referencing putative enhancers against the FANTOM5 enhancer database increases confidence in the calls.

```
epiRomics_putative_enhancers_filtered_fantom <-
  epiRomics_enhancers_filter(
    epiRomics_putative_enhancers,
    epiRomics_dB,
    epiRomics_type = "hg38_custom_fantom")

head(as.data.frame(
  epiRomics_putative_enhancers_filtered_fantom@annotations), 5)
#>   seqnames   start     end width strand
#> 1     chr1 1020834 1022106  1273      *
#> 2     chr1 1079404 1080628  1225      *
#> 3     chr1 1125003 1125650   648      *
#> 4     chr1 1127553 1128581  1029      *
#> 5     chr1 1407793 1408358   566      *
```

Human Islet Regulome Active Enhancers

```
epiRomics_putative_enhancers_filtered_regulome_active <-
  epiRomics_enhancers_filter(
    epiRomics_putative_enhancers,
    epiRomics_dB,
    epiRomics_type = "hg38_custom_regulome_active")

head(as.data.frame(
  epiRomics_putative_enhancers_filtered_regulome_active@annotations), 5)
#>   seqnames   start     end width strand
#> 1     chr1 1020834 1022106  1273      *
#> 2     chr1 1063785 1064287   503      *
#> 3     chr1 1068081 1068951   871      *
#> 4     chr1 1069171 1069416   246      *
#> 5     chr1 1069419 1069733   315      *
```

Human Islet Regulome Super Enhancers

```
epiRomics_putative_enhancers_filtered_regulome_super <-
  epiRomics_enhancers_filter(
    epiRomics_putative_enhancers,
    epiRomics_dB,
    epiRomics_type = "hg38_custom_regulome_super")

head(as.data.frame(
  epiRomics_putative_enhancers_filtered_regulome_super@annotations), 5)
```

```
#>   seqnames   start     end width strand
#> 1     chr1  941074  941572   499      *
#> 2     chr1  941703  941976   274      *
#> 3     chr1  942153  942691   539      *
#> 4     chr1  942923  943021    99      *
#> 5     chr1 7574003 7575449  1447      *
```

Ultra-Conserved Non-Coding Elements (UCNEs)

```
epiRomics_putative_enhancers_filtered_ucnes <-
  epiRomics_enhancers_filter(
    epiRomics_putative_enhancers,
    epiRomics_dB,
    epiRomics_type = "hg38_custom_ucnes")

head(as.data.frame(
  epiRomics_putative_enhancers_filtered_ucnes@annotations), 5)
#>   seqnames    start      end width strand
#> 1     chr1 10697785 10698254   470      *
#> 2     chr1 10791345 10791960   616      *
#> 3     chr1 26698321 26699133   813      *
#> 4     chr1 26723825 26724726   902      *
#> 5     chr1 61244005 61244524   520      *
```

Screening for high TF co-binding sites (enhanceosomes)

Enhancers can be redundant and not all play an active regulatory role. By intersecting putative enhancer calls with all available ChIP-seq data, we identify enhanceosome regions with high transcription factor co-binding. Results are sorted by co-binding count.

```
epiRomics_putative_enhanceosome_fantom <-
  epiRomics_enhanceosome(
    epiRomics_putative_enhancers_filtered_fantom,
    epiRomics_dB)
#>
#> >> preparing features information...      2026-03-14 17:26:54
#> >> Using Genome: hg38 ...
#> >> identifying nearest features...        2026-03-14 17:26:55
#> >> calculating distance from peak to TSS...   2026-03-14 17:26:55
#> >> assigning genomic annotation...        2026-03-14 17:26:55
#> >> Using Genome: hg38 ...
#> >> Using Genome: hg38 ...
#> >> adding gene annotation...          2026-03-14 17:27:18
#> 'select()' returned 1:many mapping between keys and columns
#> >> assigning chromosome lengths           2026-03-14 17:27:18
#> >> done...                     2026-03-14 17:27:18

head(as.data.frame(
  epiRomics_putative_enhanceosome_fantom@annotations), 5)
#>   seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
```

```
#> 1    chr20  31703684  31714653 10970      *     3    3      4      2
#> 2     chr5  95725871  95730870  5000      *     3    3      3      3
#> 3     chr3  43179244  43181850  2607      *     1    2      3      2
#> 4     chr5 139652270 139672126 19857      *     1    1      7      1
#> 5    chr10 125146386 125156982 10597      *     3    1      4      0
#>   pdx1 ChIP_Hits
#> 1    2        14
#> 2    1        13
#> 3    3        11
#> 4    1        11
#> 5    3        11
#>                                                   annotation geneChr
#> 1         Intron (ENST00000412972.2/105372589, intron 3 of 3)      20
#> 2                                              Promoter (<=1kb)       5
#> 3 Intron (ENST00000730138.1/ENST00000730138.1, intron 1 of 2)       3
#> 4                                              Promoter (<=1kb)       5
#> 5                                              Promoter (1-2kb)      10
#>   geneStart   geneEnd geneLength geneStrand    geneId
#> 1  31664645  31719417      54773          2       598
#> 2  95730872  95748441      17570          1     22836
#> 3  43151607  43153213       1607          1 105377051
#> 4 139667824 139670094       2271          2 124901084
#> 5 124989225 125158716     169492          2      1488
#>        transcriptId distanceToTSS         ENSEMBL       SYMBOL
#> 1 ENST00000677494.1          4764 ENSG00000171552       BCL2L1
#> 2 ENST00000506959.5            -2 ENSG00000164292      RHOBTB3
#> 3 ENST00000722991.1         27637            <NA> LOC105377051
#> 4 ENST00000756911.1             0            <NA> LOC124901084
#> 5 ENST00000531469.5          1734 ENSG00000175029        CTBP2
#>                              GENENAME
#> 1                          BCL2 like 1
#> 2 Rho related BTB domain containing 3
#> 3        uncharacterized LOC105377051
#> 4        uncharacterized LOC124901084
#> 5        C-terminal binding protein 2
```

```
## Evaluate calls on chromosome 1
fantom_df <- as.data.frame(
  epiRomics_putative_enhanceosome_fantom@annotations)
head(fantom_df[fantom_df$seqnames == "chr1", ], 5)
#>    seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 8      chr1 205312504 205319760  7257      *     3    2      4      1
#> 13     chr1  22412898  22415136  2239      *     1    2      2      2
#> 14     chr1 154417790 154421594  3805      *     2    2      2      2
#> 15     chr1 156501619 156505087  3469      *     2    1      4      1
#> 16     chr1 203285987 203290490  4504      *     2    1      3      1
#>    pdx1 ChIP_Hits                                   annotation
#> 8     0        10                              Promoter (1-2kb)
#> 13    2         9                             Distal Intergenic
#> 14    1         9 Intron (ENST00000622330.5/3570, intron 1 of 6)
#> 15    1         9                              Promoter (<=1kb)
#> 16    2         9                              Promoter (<=1kb)
#>    geneChr geneStart   geneEnd geneLength geneStrand    geneId
```

```
#> 8        1 205302063 205321745      19683          2     81788
#> 13       1  22416412  22418931       2520          2 105376856
#> 14       1 154429343 154449979      20637          1      3570
#> 15       1 156483261 156500779      17519          2      4209
#> 16       1 203286470 203288805       2336          1 100506775
#>         transcriptId distanceToTSS         ENSEMBL       SYMBOL
#> 8  ENST00000367157.6          1985 ENSG00000163545        NUAK2
#> 13 ENST00000767037.1          3795 ENSG00000299881 LOC105376856
#> 14 ENST00000476006.5         -7749 ENSG00000160712         IL6R
#> 15 ENST00000489057.1          -840 ENSG00000116604        MEF2D
#> 16 ENST00000779136.1             0 ENSG00000231507    LINC01353
#>                                      GENENAME
#> 8                         NUAK family kinase 2
#> 13                uncharacterized LOC105376856
#> 14                       interleukin 6 receptor
#> 15                  myocyte enhancer factor 2D
#> 16 long intergenic non-protein coding RNA 1353

## Total enhanceosome regions
length(epiRomics_putative_enhanceosome_fantom@annotations)
#> [1] 8944
```

Transcription factor co-binding analysis

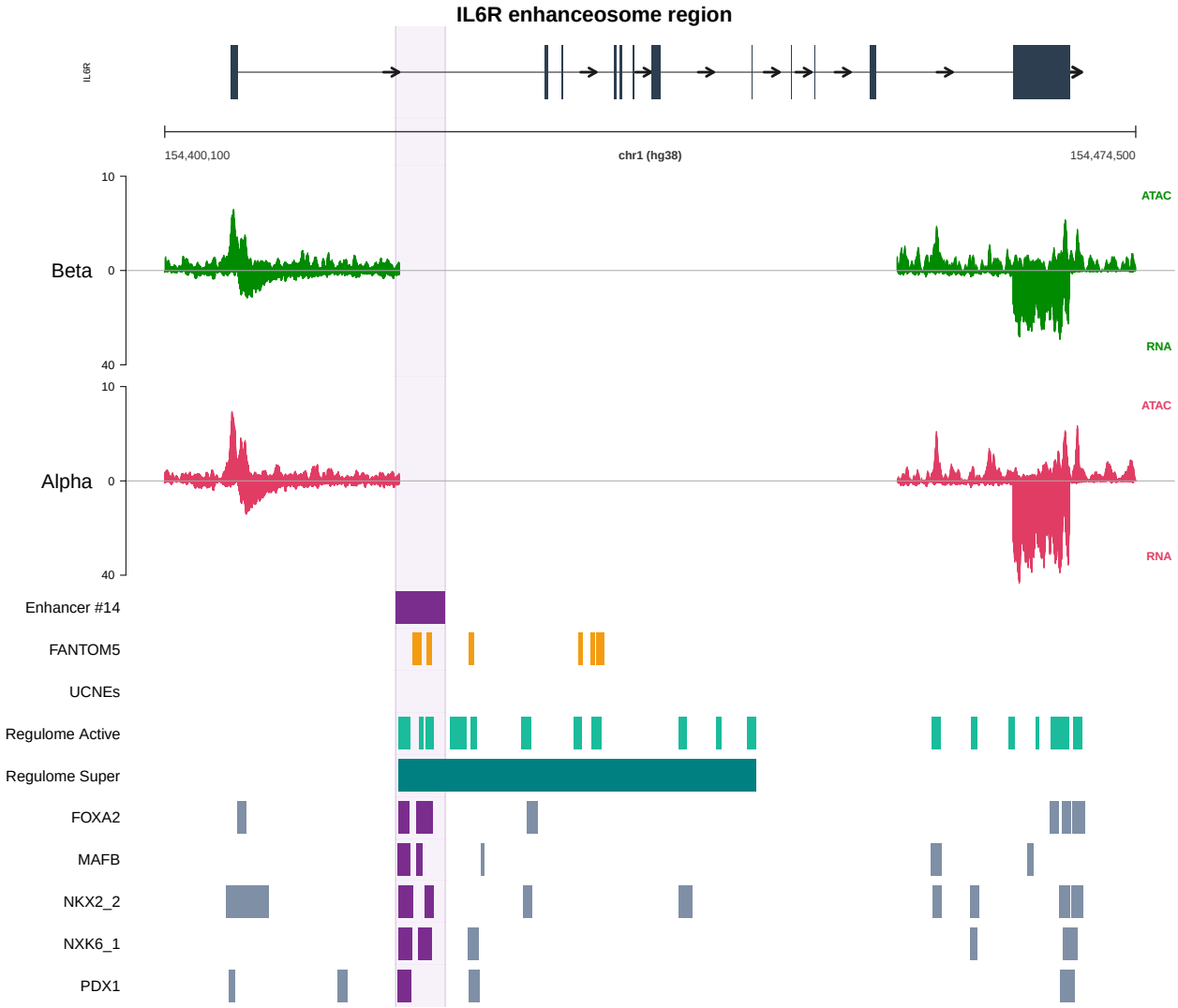
`epiRomics_tf_cobinding()` uses Fisher's exact test to identify statistically significant TF co-binding pairs within enhanceosome regions, reporting odds ratios and pointwise mutual information (PMI).

```
tf_cobinding <- epiRomics_tf_cobinding(
  epiRomics_putative_enhanceosome_fantom,
  epiRomics_dB)

tf_cobinding$pairwise
#>       tf1    tf2 n_both n_tf1_only n_tf2_only n_neither odds_ratio
#> 1  nxk6_1   pdx1    557        223        617      7547    30.5213
#> 2  nkx2_2 nxk6_1    724       2767         56      5397    25.2097
#> 3  nkx2_2   pdx1   1051       2440        123      5330    18.6645
#> 4   foxa2 nxk6_1    615       1813        165      6351    13.0475
#> 5    mafb nkx2_2    947        214       2544      5239     9.1106
#> 6   foxa2   pdx1    811       1617        363      6153     8.4986
#> 7   foxa2 nkx2_2   1771        657       1720      4796     7.5134
#> 8    mafb nxk6_1    347        814        433      7350     7.2325
#> 9    mafb   pdx1    473        688        701      7082     6.9446
#> 10  foxa2   mafb    696       1732        465      6051     5.2280
#>    pvalue    pmi fdr significant
#> 1       0 2.4437   0        TRUE
#> 2       0 1.2498   0        TRUE
#> 3       0 1.1976   0        TRUE
#> 4       0 1.5383   0        TRUE
#> 5       0 1.0633   0        TRUE
#> 6       0 1.3475   0        TRUE
#> 7       0 0.9021   0        TRUE
#> 8       0 1.7770   0        TRUE
```

```
#> 9        0 1.6340   0         TRUE
#> 10       0 1.1429   0         TRUE
```

Visualizing enhanceosome regions with signal tracks

```
epiRomics_track_layer(
  epiRomics_putative_enhanceosome_fantom,
  epiRomics_index = which(names(
    epiRomics_putative_enhanceosome_fantom@annotations) == "14"),
  epiRomics_dB = epiRomics_dB,
  epiRomics_track_connection = epiRomics_track_connection)
```

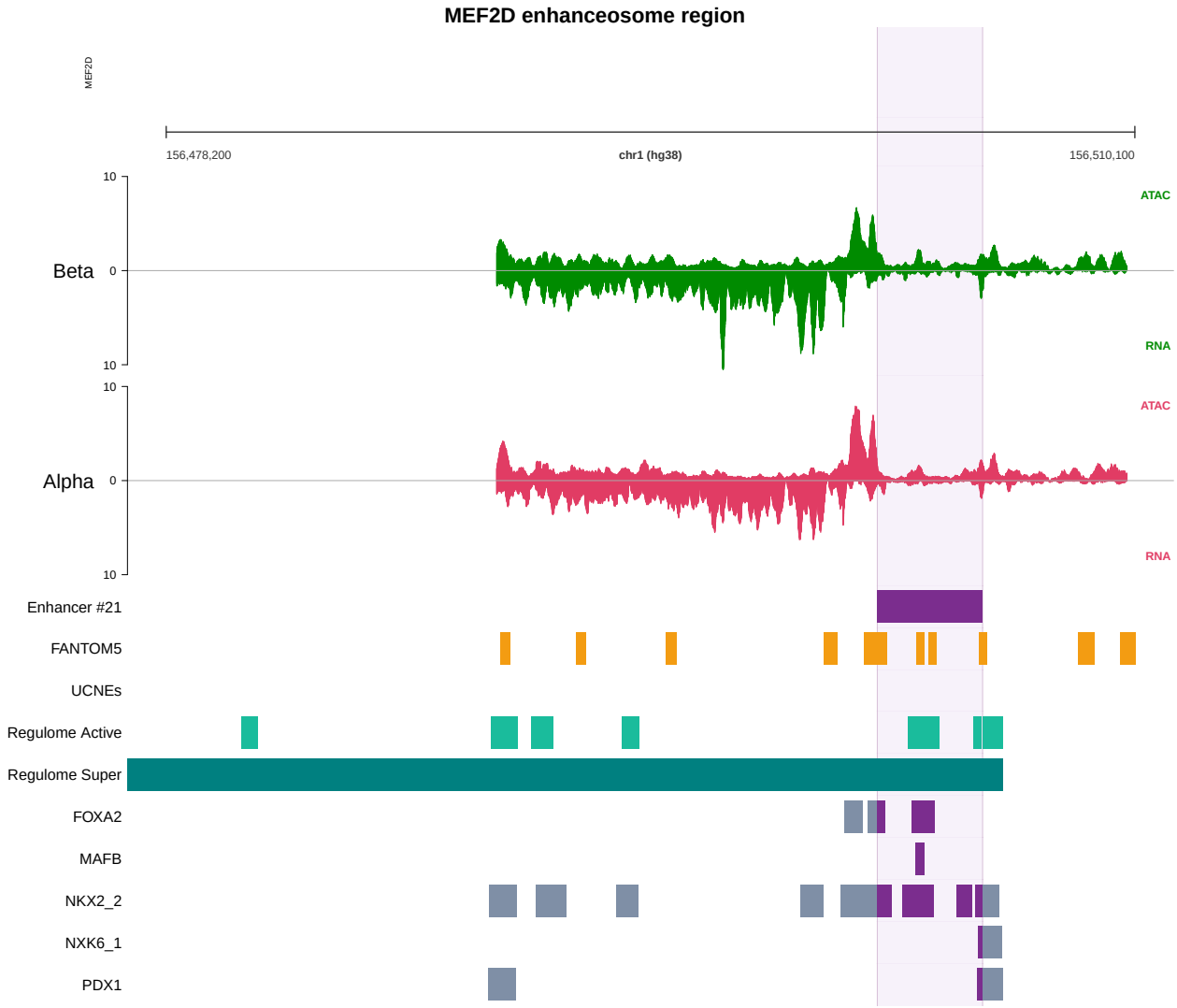


Regulome active-filtered enhanceosomes

```
epiRomics_putative_enhanceosome_regulome_active <-
  epiRomics_enhanceosome(
    epiRomics_putative_enhancers_filtered_regulome_active,
    epiRomics_dB)
#> >> preparing features information...      2026-03-14 17:27:27
#> >> Using Genome: hg38 ...
#> >> identifying nearest features...        2026-03-14 17:27:27
#> >> calculating distance from peak to TSS...   2026-03-14 17:27:27
#> >> assigning genomic annotation...        2026-03-14 17:27:27
#> >> Using Genome: hg38 ...
#> >> Using Genome: hg38 ...
#> >> adding gene annotation...          2026-03-14 17:27:31
#> 'select()' returned 1:many mapping between keys and columns
#> >> assigning chromosome lengths           2026-03-14 17:27:31
#> >> done...                     2026-03-14 17:27:31

head(as.data.frame(
  epiRomics_putative_enhanceosome_regulome_active@annotations), 5)
#>   seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 1    chr20  31703684  31714653 10970      *     3    3      4      2
#> 2     chr5  95725871  95730870  5000      *     3    3      3      3
#> 3     chr5 125408789 125412655  3867      *     3    3      2      2
#> 4     chr9  95017457  95035044 17588      *     4    1      5      1
#> 5     chr8  28363281  28370868  7588      *     2    2      3      2
#>   pdx1 ChIP_Hits
#> 1    2        14
#> 2    1        13
#> 3    3        13
#> 4    2        13
#> 5    3        12
#>                                                 annotation geneChr
#> 1     Intron (ENST00000412972.2/105372589, intron 3 of 3)      20
#> 2                                       Promoter (<=1kb)       5
#> 3 Exon (ENST00000825456.1/ENST00000825456.1, exon 2 of 3)       5
#> 4                                       Promoter (<=1kb)       9
#> 5                                                 5' UTR       8
#>   geneStart   geneEnd geneLength geneStrand    geneId
#> 1  31664645  31719417      54773          2       598
#> 2  95730872  95748441      17570          1     22836
#> 3 125358095 125601432     243338          1 101927460
#> 4  95018891  95022285       3395          1     84909
#> 5  28356690  28375379      18690          2     55893
#>        transcriptId distanceToTSS         ENSEMBL    SYMBOL
#> 1 ENST00000677494.1          4764 ENSG00000171552    BCL2L1
#> 2 ENST00000506959.5            -2 ENSG00000164292   RHOBTB3
#> 3 ENST00000825647.1         50694 ENSG00000260192 LINC02240
#> 4 ENST00000482056.2             0 ENSG00000148120     AOPEP
#> 5 ENST00000521912.1          4511 ENSG00000186918    ZNF395
#>                                     GENENAME
#> 1                                 BCL2 like 1
#> 2         Rho related BTB domain containing 3
```

```
#> 3 long intergenic non-protein coding RNA 2240
#> 4                  aminopeptidase O (putative)
#> 5                      zinc finger protein 395

reg_active_df <- as.data.frame(
  epiRomics_putative_enhanceosome_regulome_active@annotations)
head(reg_active_df[reg_active_df$seqnames == "chr1", ], 5)
#>    seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 11     chr1 205312504 205319760  7257      *     3    2      4      1
#> 18     chr1   9433722   9443473  9752      *     3    3      2      1
#> 19     chr1  22412898  22415136  2239      *     1    2      2      2
#> 20     chr1 154417790 154421594  3805      *     2    2      2      2
#> 21     chr1 156501619 156505087  3469      *     2    1      4      1
#>    pdx1 ChIP_Hits                                  annotation
#> 11    0        10                              Promoter (1-2kb)
#> 18    0         9                              Promoter (<=1kb)
#> 19    2         9                             Distal Intergenic
#> 20    1         9 Intron (ENST00000622330.5/3570, intron 1 of 6)
#> 21    1         9                              Promoter (<=1kb)
#>    geneChr geneStart   geneEnd geneLength geneStrand    geneId
#> 11       1 205302063 205321745      19683          2     81788
#> 18       1   9426649   9440564      13916          2 100506022
#> 19       1  22416412  22418931       2520          2 105376856
#> 20       1 154429343 154449979      20637          1      3570
#> 21       1 156483261 156500779      17519          2      4209
#>         transcriptId distanceToTSS         ENSEMBL       SYMBOL
#> 11 ENST00000367157.6          1985 ENSG00000163545        NUAK2
#> 18 ENST00000641940.1             0 ENSG00000284693    LINC02606
#> 19 ENST00000767037.1          3795 ENSG00000299881 LOC105376856
#> 20 ENST00000476006.5         -7749 ENSG00000160712         IL6R
#> 21 ENST00000489057.1          -840 ENSG00000116604        MEF2D
#>                                     GENENAME
#> 11                        NUAK family kinase 2
#> 18 long intergenic non-protein coding RNA 2606
#> 19                uncharacterized LOC105376856
#> 20                      interleukin 6 receptor
#> 21                  myocyte enhancer factor 2D

length(
  epiRomics_putative_enhanceosome_regulome_active@annotations)
#> [1] 15598
```

```
epiRomics_track_layer(
  epiRomics_putative_enhanceosome_regulome_active,
  epiRomics_index = which(names(
    epiRomics_putative_enhanceosome_regulome_active@annotations) == "21"),
  epiRomics_dB = epiRomics_dB,
  epiRomics_track_connection = epiRomics_track_connection)
```



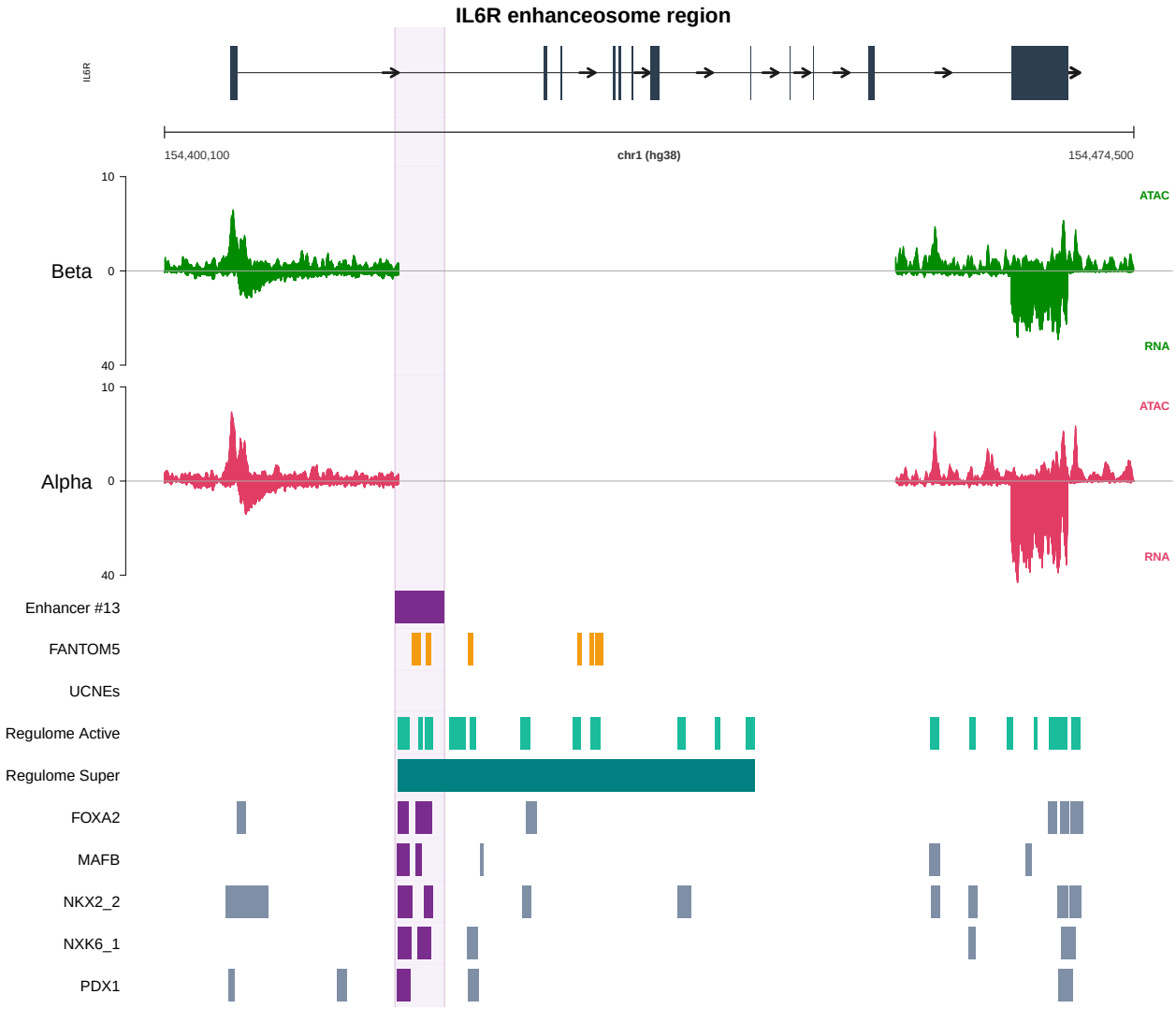
Regulome super-enhancer filtered enhanceosomes

```
epiRomics_putative_enhanceosome_regulome_super <-
  epiRomics_enhanceosome(
    epiRomics_putative_enhancers_filtered_regulome_super,
    epiRomics_dB)
#> >> preparing features information...      2026-03-14 17:27:35
#> >> Using Genome: hg38 ...
#> >> identifying nearest features...        2026-03-14 17:27:35
#> >> calculating distance from peak to TSS...   2026-03-14 17:27:36
#> >> assigning genomic annotation...        2026-03-14 17:27:36
#> >> Using Genome: hg38 ...
#> >> Using Genome: hg38 ...
#> >> adding gene annotation...          2026-03-14 17:27:40
#> 'select()' returned 1:many mapping between keys and columns
#> >> assigning chromosome lengths           2026-03-14 17:27:40
#> >> done...                     2026-03-14 17:27:40
```

```r
head(as.data.frame(
  epiRomics_putative_enhanceosome_regulome_super@annotations), 5)
#>   seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 1    chr20  31703684  31714653 10970      *     3    3      4      2
#> 2     chr5  95725871  95730870  5000      *     3    3      3      3
#> 3     chr5 125408789 125412655  3867      *     3    3      2      2
#> 4     chr9  95017457  95035044 17588      *     4    1      5      1
#> 5     chr8  28363281  28370868  7588      *     2    2      3      2
#>   pdx1 ChIP_Hits
#> 1    2        14
#> 2    1        13
#> 3    3        13
#> 4    2        13
#> 5    3        12
#>                                                 annotation geneChr
#> 1     Intron (ENST00000412972.2/105372589, intron 3 of 3)      20
#> 2                                          Promoter (<=1kb)       5
#> 3 Exon (ENST00000825456.1/ENST00000825456.1, exon 2 of 3)       5
#> 4                                          Promoter (<=1kb)       9
#> 5                                                   5' UTR       8
#>   geneStart   geneEnd geneLength geneStrand    geneId
#> 1  31664645  31719417      54773          2       598
#> 2  95730872  95748441      17570          1     22836
#> 3 125358095 125601432     243338          1 101927460
#> 4  95018891  95022285       3395          1     84909
#> 5  28356690  28375379      18690          2     55893
#>        transcriptId distanceToTSS         ENSEMBL    SYMBOL
#> 1 ENST00000677494.1          4764 ENSG00000171552    BCL2L1
#> 2 ENST00000506959.5            -2 ENSG00000164292   RHOBTB3
#> 3 ENST00000825647.1         50694 ENSG00000260192 LINC02240
#> 4 ENST00000482056.2             0 ENSG00000148120     AOPEP
#> 5 ENST00000521912.1          4511 ENSG00000186918    ZNF395
#>                                     GENENAME
#> 1                                  BCL2 like 1
#> 2         Rho related BTB domain containing 3
#> 3 long intergenic non-protein coding RNA 2240
#> 4                 aminopeptidase O (putative)
#> 5                     zinc finger protein 395

reg_super_df <- as.data.frame(
  epiRomics_putative_enhanceosome_regulome_super@annotations)
head(reg_super_df[reg_super_df$seqnames == "chr1", ], 5)
#>    seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 12     chr1  22412898  22415136  2239      *     1    2      2      2
#> 13     chr1 154417790 154421594  3805      *     2    2      2      2
#> 14     chr1 156501619 156505087  3469      *     2    1      4      1
#> 15     chr1 203285987 203290490  4504      *     2    1      3      1
#> 27     chr1  15196442  15201990  5549      *     3    0      3      1
#>    pdx1 ChIP_Hits                                   annotation
#> 12    2         9                                Distal Intergenic
#> 13    1         9  Intron (ENST00000622330.5/3570, intron 1 of 6)
#> 14    1         9                                Promoter (<=1kb)
#> 15    2         9                                Promoter (<=1kb)
```

```
#> 27    1         8 Intron (ENST00000428417.5/55092, intron 1 of 2)
#>    geneChr geneStart   geneEnd geneLength geneStrand    geneId
#> 12       1  22416412  22418931       2520          2 105376856
#> 13       1 154429343 154449979      20637          1      3570
#> 14       1 156483261 156500779      17519          2      4209
#> 15       1 203286470 203288805       2336          1 100506775
#> 27       1  15164191  15171666       7476          2    727684
#>         transcriptId distanceToTSS         ENSEMBL       SYMBOL
#> 12 ENST00000767037.1          3795 ENSG00000299881 LOC105376856
#> 13 ENST00000476006.5         -7749 ENSG00000160712         IL6R
#> 14 ENST00000489057.1          -840 ENSG00000116604        MEF2D
#> 15 ENST00000779136.1             0 ENSG00000231507    LINC01353
#> 27 ENST00000832098.1        -24776 ENSG00000204464   TMEM51-AS2
#>                                        GENENAME
#> 12                uncharacterized LOC105376856
#> 13                      interleukin 6 receptor
#> 14                  myocyte enhancer factor 2D
#> 15 long intergenic non-protein coding RNA 1353
#> 27                      TMEM51 antisense RNA 2

length(
  epiRomics_putative_enhanceosome_regulome_super@annotations)
#> [1] 4867
```

```
epiRomics_track_layer(
  epiRomics_putative_enhanceosome_regulome_super,
  epiRomics_index = which(names(
    epiRomics_putative_enhanceosome_regulome_super@annotations) == "13"),
  epiRomics_dB = epiRomics_dB,
  epiRomics_track_connection = epiRomics_track_connection)
```



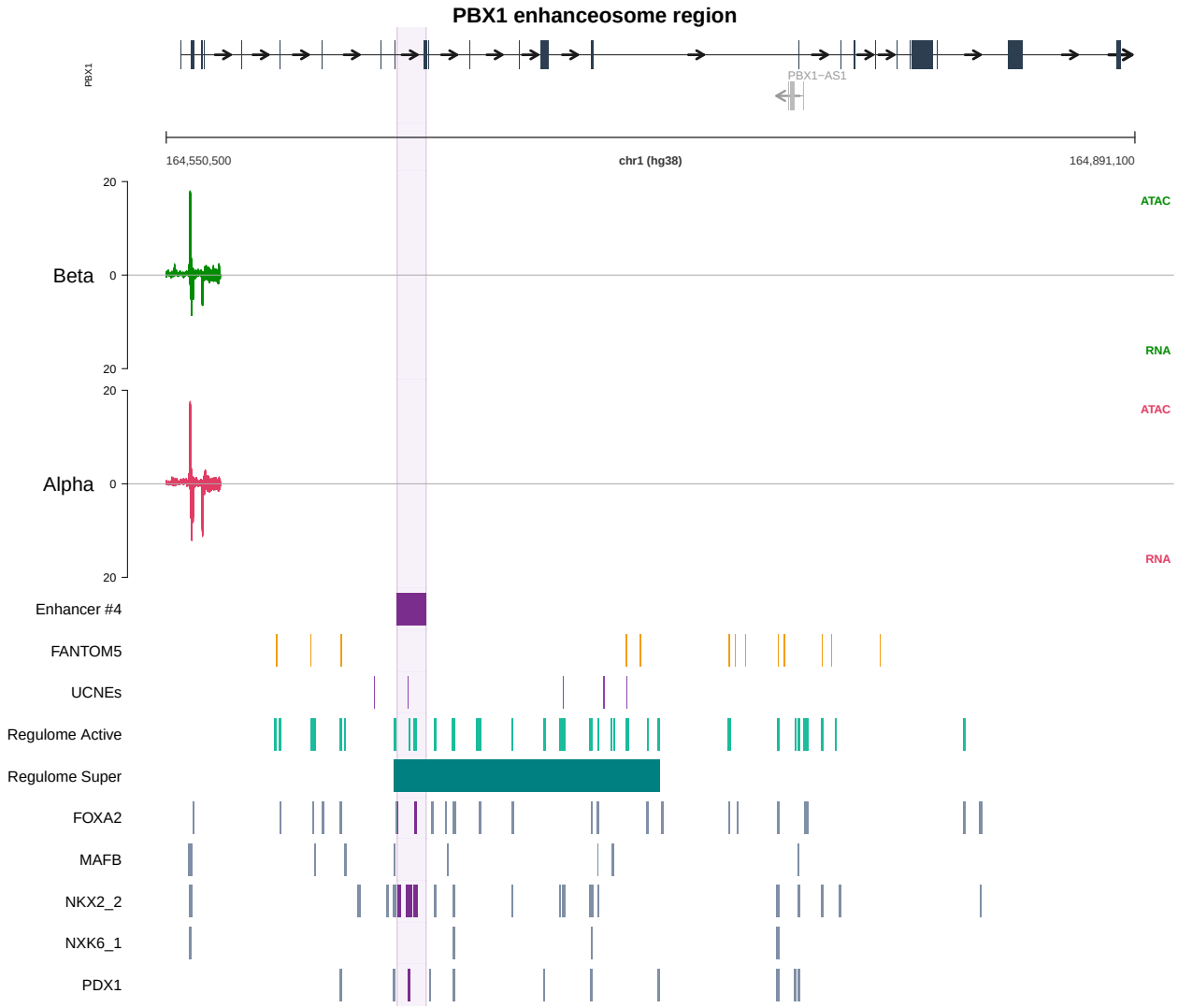
UCNE-filtered enhanceosomes

```
epiRomics_putative_enhanceosome_ucnes <-
  epiRomics_enhanceosome(
    epiRomics_putative_enhancers_filtered_ucnes,
    epiRomics_dB)
#> >> preparing features information...      2026-03-14 17:27:47
#> >> Using Genome: hg38 ...
#> >> identifying nearest features...        2026-03-14 17:27:47
#> >> calculating distance from peak to TSS...   2026-03-14 17:27:47
#> >> assigning genomic annotation...        2026-03-14 17:27:47
#> >> Using Genome: hg38 ...
#> >> Using Genome: hg38 ...
#> >> adding gene annotation...          2026-03-14 17:27:51
#> 'select()' returned 1:many mapping between keys and columns
#> >> assigning chromosome lengths           2026-03-14 17:27:51
#> >> done...                     2026-03-14 17:27:51
```

```r
head(as.data.frame(
  epiRomics_putative_enhanceosome_ucnes@annotations), 5)
#>   seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 1     chr5 125408789 125412655  3867      *     3    3      2      2
#> 2    chr10  61837529  61841040  3512      *     2    2      2      2
#> 3    chr18  75212524  75215519  2996      *     2    0      3      2
#> 4     chr1 164631705 164641897 10193      *     2    0      4      0
#> 5     chr7 114412926 114414663  1738      *     2    0      1      2
#>   pdx1 ChIP_Hits
#> 1    3        13
#> 2    2        10
#> 3    1         8
#> 4    1         7
#> 5    2         7
#>                                                 annotation geneChr
#> 1 Exon (ENST00000825456.1/ENST00000825456.1, exon 2 of 3)       5
#> 2         Exon (ENST00000717541.1/107984237, exon 4 of 9)      10
#> 3                                          Promoter (1-2kb)      18
#> 4                                          Promoter (<=1kb)       1
#> 5                                          Promoter (<=1kb)       7
#>   geneStart   geneEnd geneLength geneStrand    geneId
#> 1 125358095 125601432     243338          1 101927460
#> 2  61781199  61830884      49686          2 107984237
#> 3  75210797  75289944      79148          1     10194
#> 4 164630981 164799889     168909          1      5087
#> 5 114414274 114690163     275890          1     93986
#>        transcriptId distanceToTSS         ENSEMBL    SYMBOL
#> 1 ENST00000825647.1         50694 ENSG00000260192 LINC02240
#> 2 ENST00000717533.1         -6645            <NA> LINC02625
#> 3 ENST00000580243.3          1727 ENSG00000179981     TSHZ1
#> 4 ENST00000482110.5           724 ENSG00000185630      PBX1
#> 5 ENST00000393494.6             0 ENSG00000128573     FOXP2
#>                                     GENENAME
#> 1 long intergenic non-protein coding RNA 2240
#> 2 long intergenic non-protein coding RNA 2625
#> 3             teashirt zinc finger homeobox 1
#> 4                              PBX homeobox 1
#> 5                             forkhead box P2

## Evaluate calls on chromosome 1
ucne_df <- as.data.frame(
  epiRomics_putative_enhanceosome_ucnes@annotations)
head(ucne_df[ucne_df$seqnames == "chr1", ], 5)
#>    seqnames     start       end width strand foxa2 mafb nkx2_2 nxk6_1
#> 4      chr1 164631705 164641897 10193      *     2    0      4      0
#> 9      chr1  10791345  10791960   616      *     1    1      1      1
#> 10     chr1  61452195  61453564  1370      *     1    2      1      0
#> 11     chr1 214001861 214003628  1768      *     1    1      1      1
#> 27     chr1  26698321  26699133   813      *     1    0      1      0
#>    pdx1 ChIP_Hits                                      annotation
#> 4     1         7                                  Promoter (<=1kb)
#> 9     1         5 Intron (ENST00000377022.8/54897, intron 1 of 20)
#> 10    1         5 Intron (ENST00000371191.5/4774, intron 10 of 10)
```

```
#> 11    1         5   Intron (ENST00000366958.9/5629, intron 2 of 4)
#> 27    1         3                                  Promoter (2-3kb)
#>    geneChr geneStart   geneEnd geneLength geneStrand geneId
#> 4        1 164630981 164799889     168909          1   5087
#> 9        1  10636604  10796646     160043          2  54897
#> 10       1  61406672  61455874      49203          1   4774
#> 11       1 213989718 213996964       7247          1   5629
#> 27       1  26696032  26731568      35537          1   8289
#>         transcriptId distanceToTSS         ENSEMBL SYMBOL
#> 4  ENST00000482110.5           724 ENSG00000185630   PBX1
#> 9  ENST00000377022.8          4686 ENSG00000130940  CASZ1
#> 10 ENST00000493627.1         45523 ENSG00000162599   NFIA
#> 11 ENST00000607425.1         12143 ENSG00000117707  PROX1
#> 27 ENST00000637465.1          2289 ENSG00000117713 ARID1A
#>                         GENENAME
#> 4                 PBX homeobox 1
#> 9           castor zinc finger 1
#> 10            nuclear factor I A
#> 11           prospero homeobox 1
#> 27 AT-rich interaction domain 1A

## Total enhanceosome regions
length(epiRomics_putative_enhanceosome_ucnes@annotations)
#> [1] 129
```

```
epiRomics_track_layer(
  epiRomics_putative_enhanceosome_ucnes,
  epiRomics_index = 4,
  epiRomics_dB = epiRomics_dB,
  epiRomics_track_connection = epiRomics_track_connection)
```



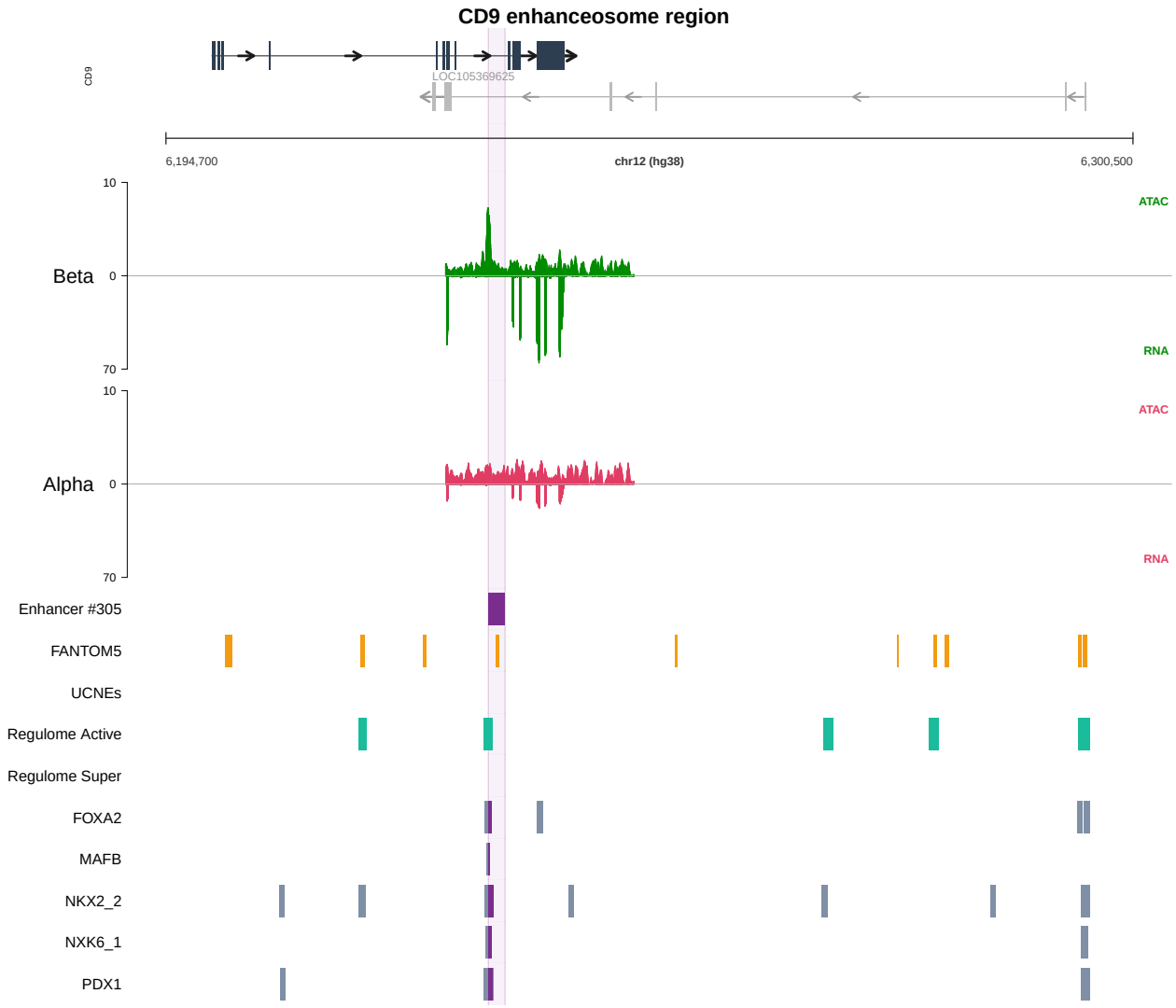
Integrating differential chromatin accessibility

How can we use putative enhanceosome regions to infer biology between cell states? Here we integrate DiffBind ATAC-seq data showing differential chromatin accessibility between alpha and beta cells.

```
b.v.a <- read.csv(
  file.path(extdata_dir,
    "DBA_Beta_Versus_Alpha.csv"))
b.v.a <- GRanges(b.v.a)

## Filter for beta-enriched chromatin regions
beta.enriched <- b.v.a[b.v.a$Fold >= 1 & b.v.a$Conc>=3, ]

## Connect to putative enhanceosomes
beta_enhancer_regions <-
  epiRomics_regions_of_interest(
    epiRomics_putative_enhanceosome_fantom,
    beta.enriched)
```

```
#> ROI filter: 61 of 8944 regions retained

head(as.data.frame(
  beta_enhancer_regions@annotations), 5)
#>     seqnames     start       end width strand foxa2 mafb nkx2_2
#> 20     chr14  76828285  76835529  7245      *     2    1      3
#> 44     chr17  59367192  59374437  7246      *     3    0      2
#> 65     chr14 100408755 100409817  1063      *     2    0      2
#> 72     chr17  59843950  59849848  5899      *     2    1      2
#> 305    chr12   6229973   6231794  1822      *     1    1      1
#>     nxk6_1 pdx1 ChIP_Hits       annotation geneChr geneStart
#> 20       1    2         9 Promoter (1-2kb)      14  76826496
#> 44       2    1         8 Promoter (1-2kb)      17  59366083
#> 65       2    1         7 Promoter (2-3kb)      14 100406599
#> 72       1    1         7 Promoter (2-3kb)      17  59841266
#> 305      1    1         5 Promoter (<=1kb)      12   6232186
#>       geneEnd geneLength geneStrand    geneId      transcriptId
#> 20   76870302      43807          1    145497 ENST00000691122.1
#> 44   59366154         72          1 100616204 ENST00000584140.1
#> 65  100408928       2330          1 105370664 ENST00000556080.2
#> 72   59841337         72          1    406991 ENST00000362134.1
#> 305   6236262       4077          1       928 ENST00000540891.1
#>     distanceToTSS         ENSEMBL       SYMBOL
#> 20           1789 ENSG00000100565      LRRC74A
#> 44           1109 ENSG00000263857      MIR4729
#> 65           2156            <NA> LOC105370664
#> 72           2684 ENSG00000284190        MIR21
#> 305          -392 ENSG00000010278          CD9
#>                              GENENAME
#> 20  leucine rich repeat containing 74A
#> 44                      microRNA 4729
#> 65        uncharacterized LOC105370664
#> 72                        microRNA 21
#> 305                      CD9 molecule
```

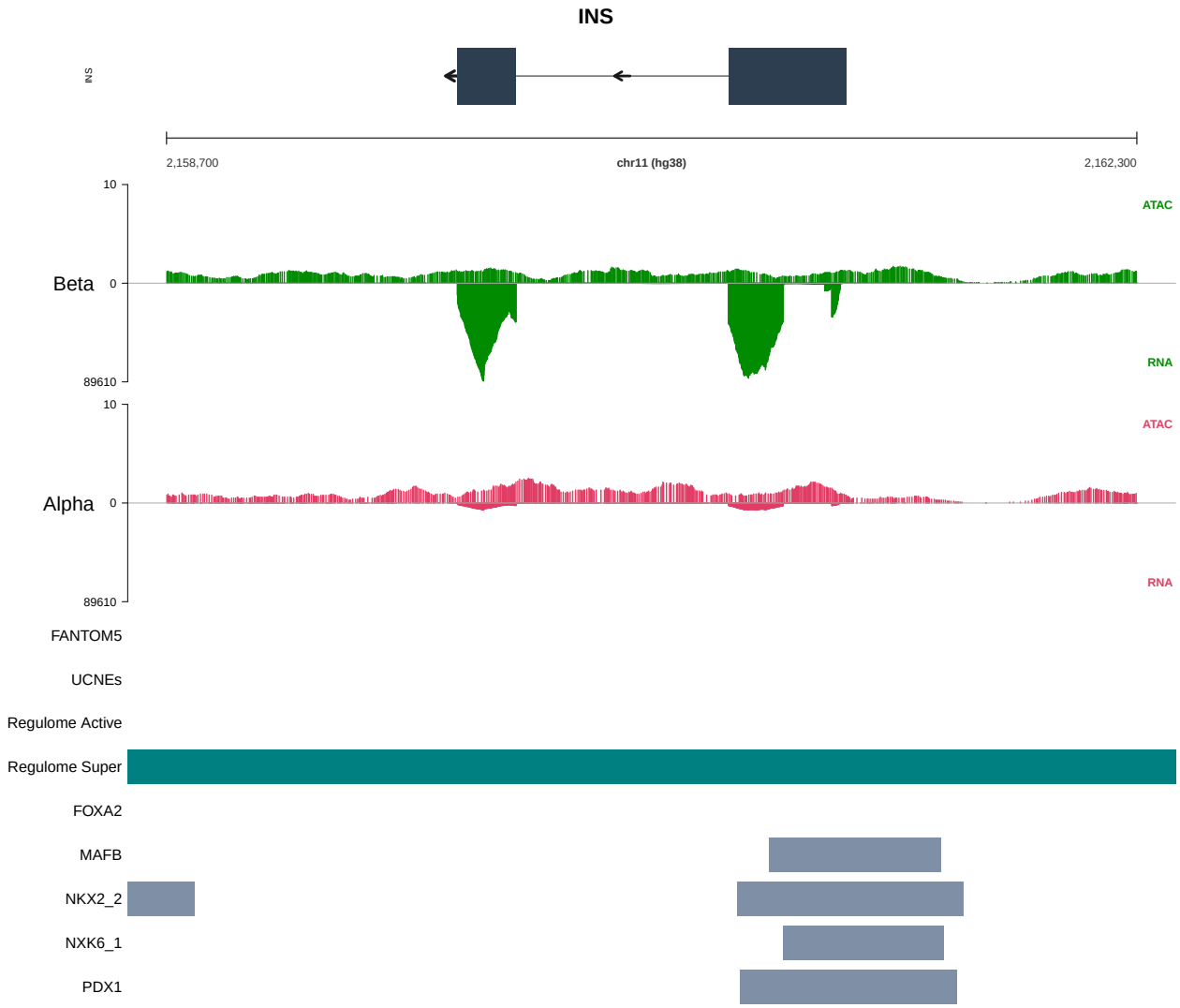
```
epiRomics_track_layer(
  beta_enhancer_regions,
  epiRomics_index = which(names(
    beta_enhancer_regions@annotations) == "305"),
  epiRomics_dB = epiRomics_dB,
  epiRomics_track_connection = epiRomics_track_connection)
```



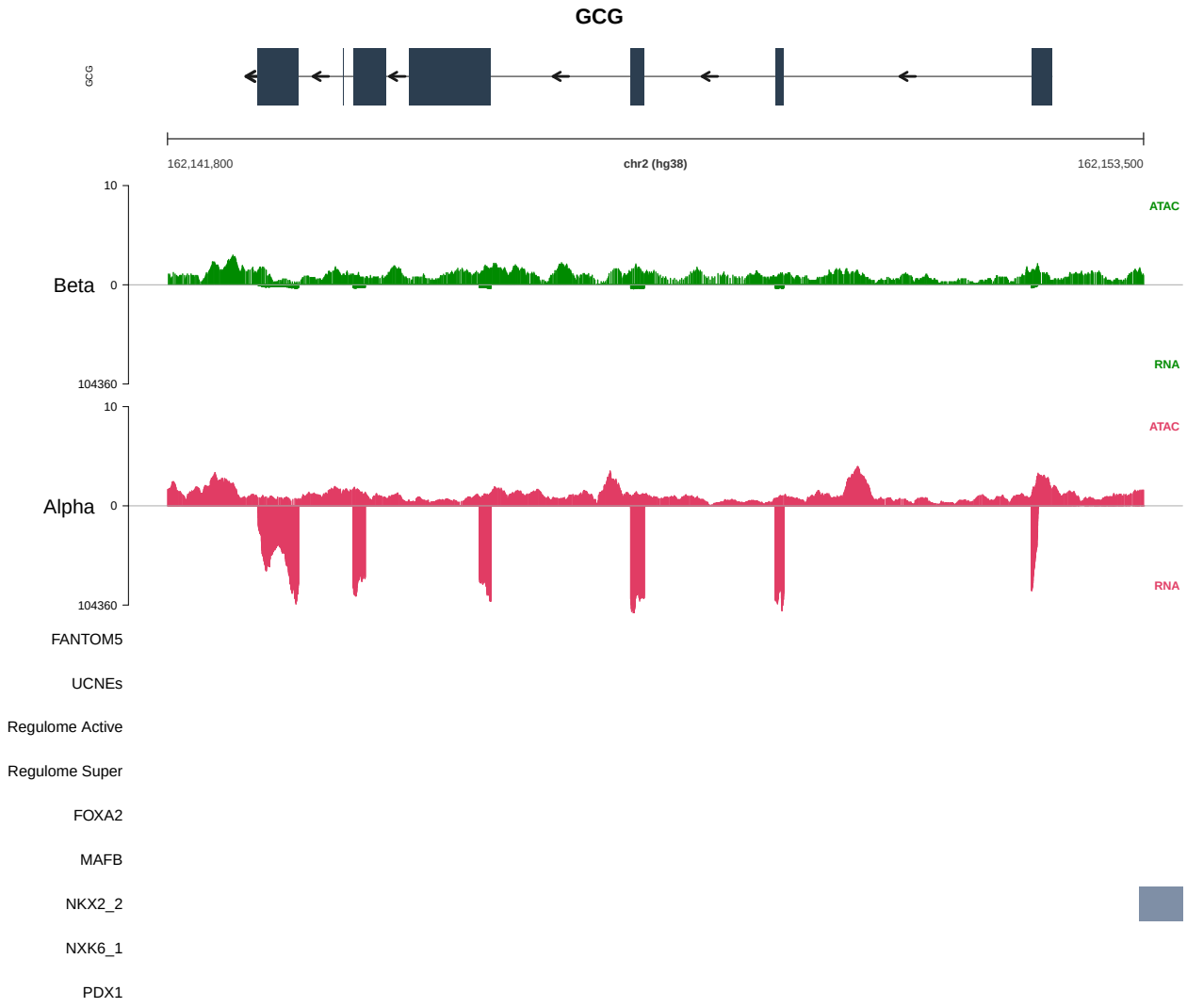
Gene-centered visualization

`epiRomics_track_layer_gene()` provides a gene-focused view combining all database tracks (histone marks, ChIP, annotations) with BigWig signal—no enhanceosome pipeline required.

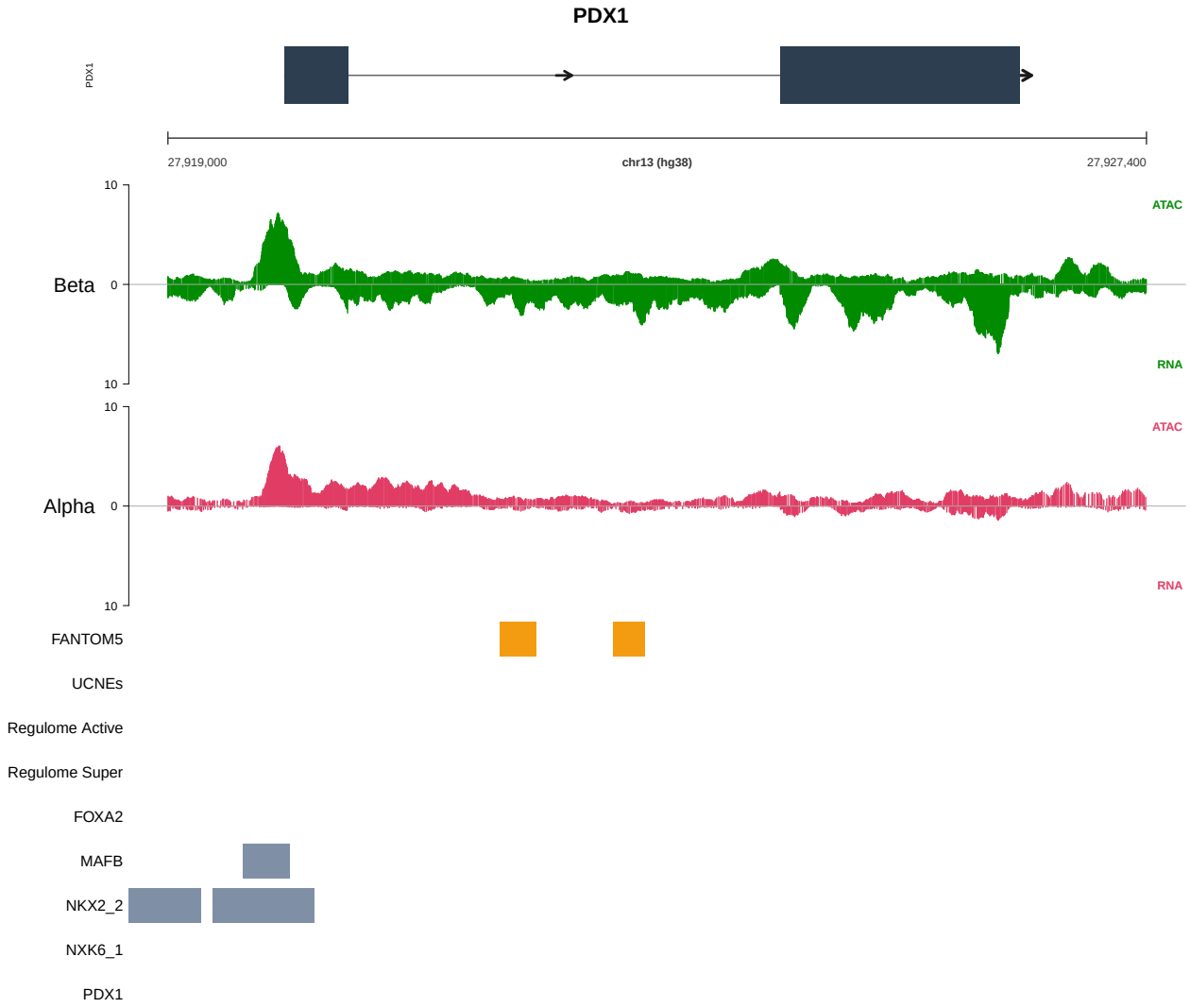
```
epiRomics_track_layer_gene("INS",
  epiRomics_dB,
  epiRomics_track_connection)
```




```
epiRomics_track_layer_gene("GCG",
  epiRomics_dB,
  epiRomics_track_connection)
```




```r
## Wider padding captures distal regulatory elements
epiRomics_track_layer_gene("PDX1",
  epiRomics_dB,
  epiRomics_track_connection,
  padding = 5000)
```



Chromatin state classification

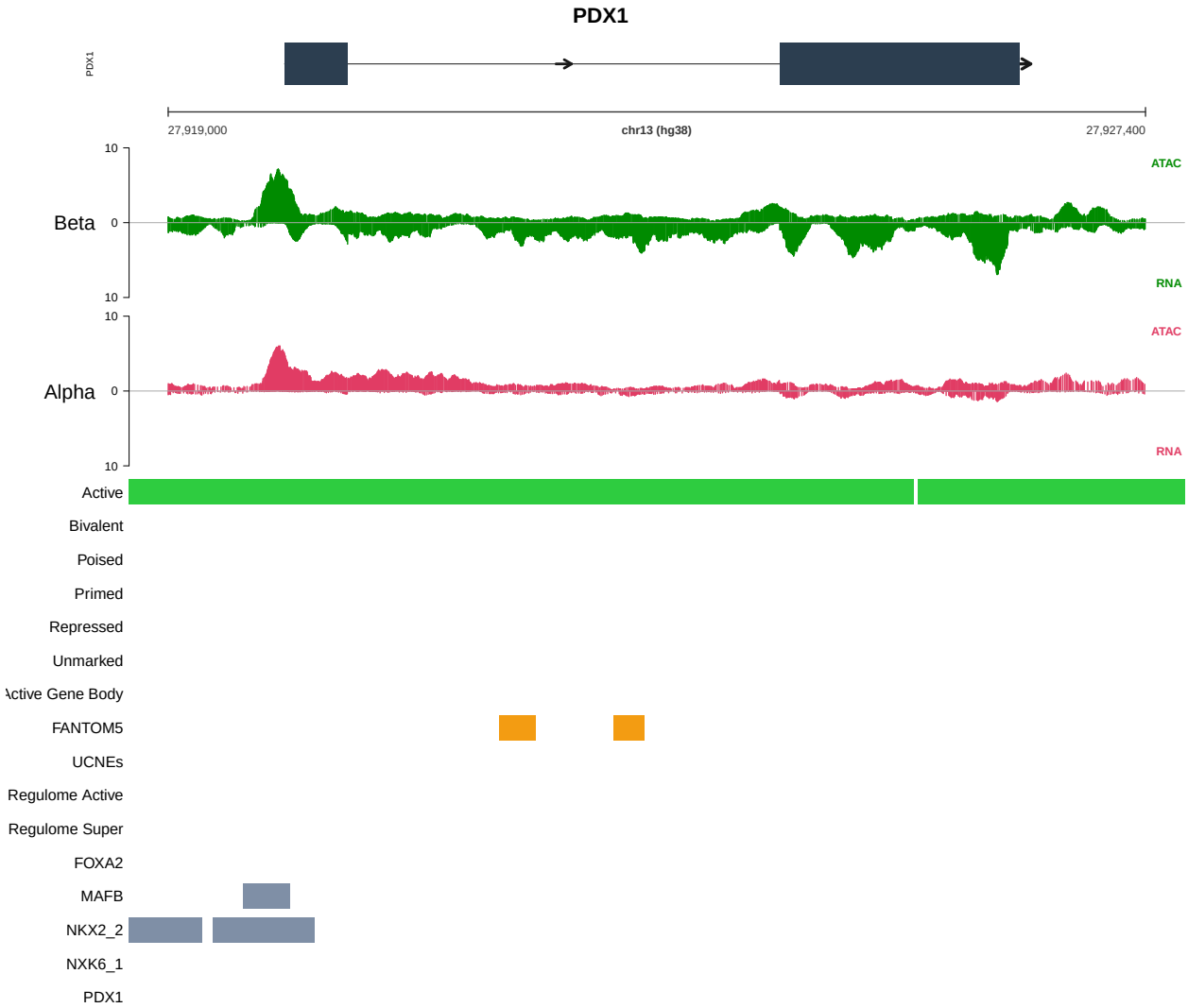
`epiRomics_chromatin_states()` classifies genomic regions by their histone mark combinations into six biologically meaningful states: active, bivalent, poised, primed, repressed, and unmarked. Classification follows ChromHMM/Roadmap Epigenomics conventions (Ernst & Kellis 2012; Kundaje et al. 2015), with TSS proximity refinement.

```
chromatin_states <-
  epiRomics_chromatin_states(epiRomics_dB)

table(chromatin_states$chromatin_state)
#>
#>    active  bivalent    poised    primed repressed  unmarked
#>     56337      1988      1591    100815    130224    153140
table(chromatin_states$genomic_context)
#>
#>  gene_body intergenic   promoter
#>     288663     123490      31942
```

```r
pdx1_states <- chromatin_states[
  chromatin_states$seqnames == "chr13" &
    chromatin_states$start >= 27900000 &
    chromatin_states$end   <= 28000000, ]
head(pdx1_states, 10)
#>        seqnames    start      end width chromatin_state
#> 300209    chr13 27901032 27901702   671          active
#> 300210    chr13 27902101 27902207   107          primed
#> 300211    chr13 27905074 27906552  1479          active
#> 300212    chr13 27906825 27907184   360          active
#> 300213    chr13 27907303 27910742  3440          active
#> 300214    chr13 27910859 27913272  2414          active
#> 300215    chr13 27915283 27916414  1132          active
#> 300216    chr13 27916540 27925404  8865          active
#> 300217    chr13 27925444 27929964  4521          active
#> 300218    chr13 27930118 27930567   450          active
#>        genomic_context                         marks_present n_marks
#> 300209       gene_body                       h3k27ac,h3k4me1       2
#> 300210       gene_body                               h3k4me1       1
#> 300211       gene_body                  h3k27ac,h3k4me1,h2az       3
#> 300212       gene_body                               h3k27ac       1
#> 300213       gene_body                       h3k27ac,h3k4me1       2
#> 300214       gene_body                       h3k27ac,h3k4me1       2
#> 300215        promoter                  h3k27ac,h3k4me1,h2az       3
#> 300216        promoter h3k27ac,h3k4me1,h3k4me3,h3k36me3,h2az       5
#> 300217       gene_body              h3k27ac,h3k4me3,h3k36me3       3
#> 300218      intergenic                               h3k27ac       1
#>        is_hotspot h2az_overlap
#> 300209      FALSE        FALSE
#> 300210      FALSE        FALSE
#> 300211       TRUE         TRUE
#> 300212      FALSE        FALSE
#> 300213      FALSE        FALSE
#> 300214      FALSE        FALSE
#> 300215       TRUE         TRUE
#> 300216       TRUE         TRUE
#> 300217       TRUE        FALSE
#> 300218      FALSE        FALSE
```

Visualize the PDX1 locus with chromatin state tracks overlaid:

```r
epiRomics_track_layer_gene("PDX1",
  epiRomics_dB,
  epiRomics_track_connection,
  chromatin_states = chromatin_states,
  show_chromatin = TRUE,
  show_enhancer_highlight = TRUE,
  padding = 5000)
```



Session Information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Sequoia 15.7.3
#>
#> Matrix products: default
#> BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versi
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
#>
#> locale:
#> [1] en_US/en_US/en_US/C/en_US/en_US
#>
#> time zone: America/Los_Angeles
#> tzcode source: internal
#>
```

```
#> attached base packages:
#> [1] stats4    stats     graphics  grDevices utils     datasets
#> [7] methods   base
#>
#> other attached packages:
#>  [1] BSgenome.Hsapiens.UCSC.hg38_1.4.5
#>  [2] BSgenome_1.78.0
#>  [3] rtracklayer_1.70.1
#>  [4] BiocIO_1.20.0
#>  [5] Biostrings_2.78.0
#>  [6] XVector_0.50.0
#>  [7] GenomeInfoDb_1.46.2
#>  [8] org.Hs.eg.db_3.22.0
#>  [9] TxDb.Hsapiens.UCSC.hg38.knownGene_3.22.0
#> [10] GenomicFeatures_1.62.0
#> [11] AnnotationDbi_1.72.0
#> [12] Biobase_2.70.0
#> [13] GenomicRanges_1.62.1
#> [14] Seqinfo_1.0.0
#> [15] IRanges_2.44.0
#> [16] S4Vectors_0.48.0
#> [17] BiocGenerics_0.56.0
#> [18] generics_0.1.4
#> [19] epiRomics_0.99.0
#>
#> loaded via a namespace (and not attached):
#>   [1] RColorBrewer_1.1-3
#>   [2] jsonlite_2.0.0
#>   [3] magrittr_2.0.4
#>   [4] ggtangle_0.1.1
#>   [5] farver_2.1.2
#>   [6] rmarkdown_2.30
#>   [7] fs_1.6.7
#>   [8] vctrs_0.7.1
#>   [9] memoise_2.0.1
#>  [10] Rsamtools_2.26.0
#>  [11] RCurl_1.98-1.17
#>  [12] ggtree_4.0.4
#>  [13] tinytex_0.58
#>  [14] htmltools_0.5.9
#>  [15] S4Arrays_1.10.1
#>  [16] TxDb.Hsapiens.UCSC.hg19.knownGene_3.22.1
#>  [17] plotrix_3.8-14
#>  [18] AnnotationHub_4.0.0
#>  [19] curl_7.0.0
#>  [20] SparseArray_1.10.9
#>  [21] gridGraphics_0.5-1
#>  [22] KernSmooth_2.23-26
#>  [23] htmlwidgets_1.6.4
#>  [24] plyr_1.8.9
#>  [25] httr2_1.2.2
#>  [26] cachem_1.1.0
#>  [27] GenomicAlignments_1.46.0
```

```
#>  [28] igraph_2.2.2
#>  [29] lifecycle_1.0.5
#>  [30] pkgconfig_2.0.3
#>  [31] Matrix_1.7-4
#>  [32] R6_2.6.1
#>  [33] fastmap_1.2.0
#>  [34] MatrixGenerics_1.22.0
#>  [35] digest_0.6.39
#>  [36] aplot_0.2.9
#>  [37] enrichplot_1.29.2.001
#>  [38] patchwork_1.3.2
#>  [39] regioneR_1.42.0
#>  [40] RSQLite_2.4.6
#>  [41] filelock_1.0.3
#>  [42] org.Mm.eg.db_3.22.0
#>  [43] httr_1.4.8
#>  [44] abind_1.4-8
#>  [45] compiler_4.5.2
#>  [46] fontquiver_0.2.1
#>  [47] bit64_4.6.0-1
#>  [48] withr_3.0.2
#>  [49] S7_0.2.1
#>  [50] BiocParallel_1.44.0
#>  [51] DBI_1.3.0
#>  [52] gplots_3.3.0
#>  [53] R.utils_2.13.0
#>  [54] MASS_7.3-65
#>  [55] ChIPseeker_1.47.1
#>  [56] rappdirs_0.3.4
#>  [57] DelayedArray_0.36.0
#>  [58] rjson_0.2.23
#>  [59] caTools_1.18.3
#>  [60] gtools_3.9.5
#>  [61] tools_4.5.2
#>  [62] ape_5.8-1
#>  [63] R.oo_1.27.1
#>  [64] glue_1.8.0
#>  [65] restfulr_0.0.16
#>  [66] nlme_3.1-168
#>  [67] GOSemSim_2.36.0
#>  [68] grid_4.5.2
#>  [69] reshape2_1.4.5
#>  [70] fgsea_1.36.2
#>  [71] gtable_0.3.6
#>  [72] tzdb_0.5.0
#>  [73] R.methodsS3_1.8.2
#>  [74] tidyr_1.3.2
#>  [75] data.table_1.18.2.1
#>  [76] hms_1.1.4
#>  [77] ggrepel_0.9.7
#>  [78] BiocVersion_3.22.0
#>  [79] pillar_1.11.1
#>  [80] stringr_1.6.0
```

```
#>  [81] yulab.utils_0.2.4
#>  [82] vroom_1.7.0
#>  [83] splines_4.5.2
#>  [84] dplyr_1.2.0
#>  [85] treeio_1.34.0
#>  [86] BiocFileCache_3.0.0
#>  [87] lattice_0.22-7
#>  [88] bit_4.6.0
#>  [89] tidyselect_1.2.1
#>  [90] fontLiberation_0.1.0
#>  [91] GO.db_3.22.0
#>  [92] knitr_1.51
#>  [93] fontBitstreamVera_0.1.1
#>  [94] TxDb.Mmusculus.UCSC.mm10.knownGene_3.10.0
#>  [95] SummarizedExperiment_1.40.0
#>  [96] xfun_0.56
#>  [97] annotatr_1.36.0
#>  [98] matrixStats_1.5.0
#>  [99] stringi_1.8.7
#> [100] UCSC.utils_1.6.1
#> [101] lazyeval_0.2.2
#> [102] ggfun_0.2.0
#> [103] yaml_2.3.12
#> [104] boot_1.3-32
#> [105] evaluate_1.0.5
#> [106] codetools_0.2-20
#> [107] cigarillo_1.0.0
#> [108] gdtools_0.5.0
#> [109] tibble_3.3.1
#> [110] qvalue_2.42.0
#> [111] BiocManager_1.30.27
#> [112] ggplotify_0.1.3
#> [113] cli_3.6.5
#> [114] systemfonts_1.3.2
#> [115] dichromat_2.0-0.1
#> [116] Rcpp_1.1.1
#> [117] dbplyr_2.5.2
#> [118] png_0.1-8
#> [119] XML_3.99-0.22
#> [120] parallel_4.5.2
#> [121] ggplot2_4.0.2
#> [122] readr_2.2.0
#> [123] blob_1.3.0
#> [124] DOSE_4.4.0
#> [125] bitops_1.0-9
#> [126] tidytree_0.4.7
#> [127] ggiraph_0.9.6
#> [128] scales_1.4.0
#> [129] purrr_1.2.1
#> [130] crayon_1.5.3
#> [131] rlang_1.1.7
#> [132] cowplot_1.2.0
#> [133] fastmatch_1.1-8
```

```
#> [134] KEGGREST_1.50.0
```